

Harajuku & Yoyogi (Washington Heights), Tokyo

Then-and-now aerial photographs, how the ground was made, and the disasters people chose to remember — compiled automatically from public Japanese government data.



原宿・代々木（東京都渋谷区）

Study area	2.2 km × 1.4 km around 35.6700° N, 139.7000° E
Oldest aerial photograph	c. 1936–1942
Aerial series in this brief	c. 1936–1942, 1945–1950, 1961–1969, 1974–1978, 1979–1983, 1984–1986, 1987–1990, 2019
Main natural landform	Mountain and hill slopes (41%)
Former river channels	none mapped
Filled, reclaimed or drained land	20%
Former sea, river or pond	none mapped
Meiji-era paddies, marsh and flats	rice paddy 13%
Disaster memorials within 3 km	1 (earthquake 1)

What this brief is. A reading aid for comparing the same ground across decades. It shows what the photographs and GSI's land surveys record; it does not describe current conditions, access or opening hours.

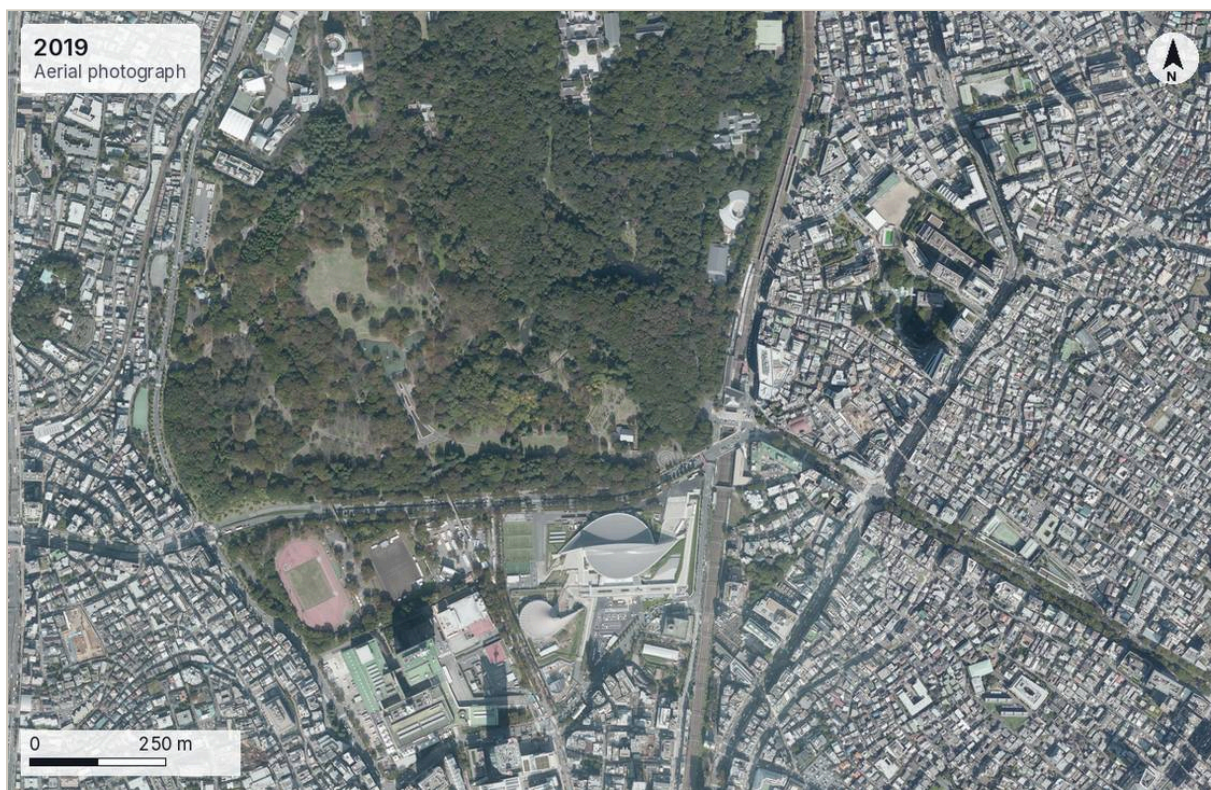
What it is not. Not a hazard assessment or a property survey. Landform classes describe typical tendencies of the ground; for present-day risk, read the official hazard map of the municipality.

Edition 2026-09-27. Aerial photographs, landform and Meiji-era lowland data: Geospatial Information Authority of Japan (GSI), processed by Japan Time Atlas. Memorial data: GSI Natural Disaster Memorials, summarised and translated by Japan Time Atlas.

The same ground, c. 1936–1942 and 2019



c. 1936–1942 aerial photograph — GSI Tiles *ort_riku10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



2019 annual orthophoto — GSI Tiles *nendophoto2019*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

The decades in between

Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



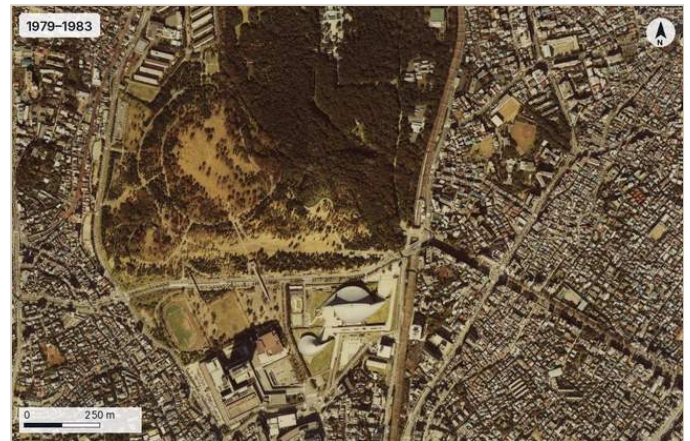
1945–1950 (US military photography) · GSI Tiles *ort_USA10*, processed



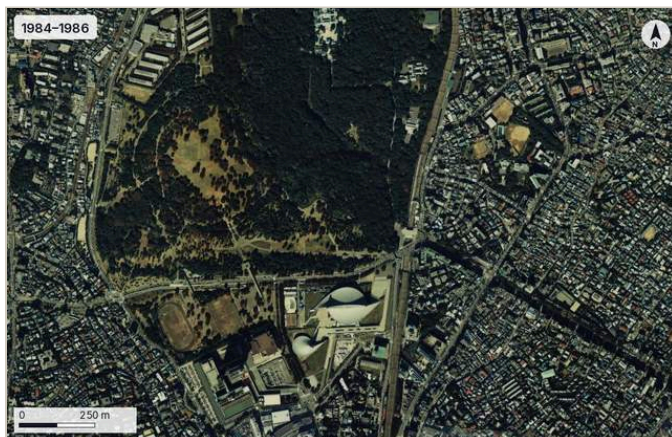
1961–1969 · GSI Tiles *ort_old10*, processed



1974–1978 · GSI Tiles *gazo1*, processed



1979–1983 · GSI Tiles *gazo2*, processed

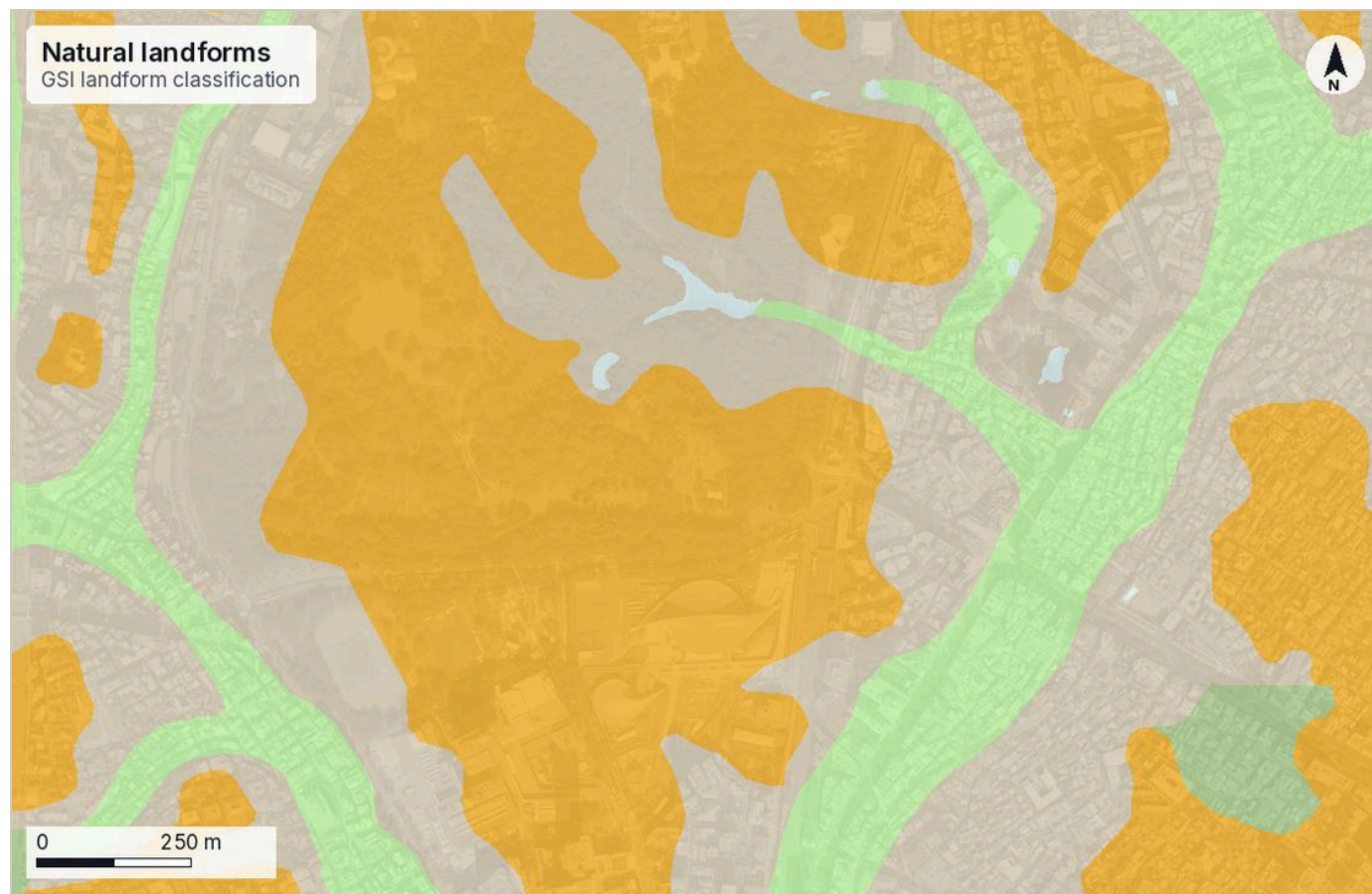


1984–1986 · GSI Tiles *gazo3*, processed



1987–1990 · GSI Tiles *gazo4*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.

- | | |
|----------------------------------|----------------------------------|
| Mountain and hill slopes 41% | Plateaus and terraces 41% |
| Floodplain and coastal plain 16% | Hollows and shallow valleys 1.5% |
| Water 0.5% | |

What the main landforms mean

Mountain and hill slopes (41%). Ridges, valleys and fairly steep ground, including terrace-scarp slopes and volcanic ground. Heavy rain or earthquakes can trigger slope failures, debris flows and landslides.

Plateaus and terraces (41%). Flat ground standing in steps above its surroundings, left behind as the land around it was eroded. River flooding is rarely a threat unless the terrace is only slightly above the river; watch the slopes at its edges. Firm ground: shaking is relatively weak and liquefaction is unlikely.

Floodplain and coastal plain (16%). Low, flat ground of little relief, built from sand and mud left by floods, or former seabed that emerged. River flooding. Ground is softer nearer the coast and shakes somewhat more; liquefaction tendency is strong on coastal plains and somewhat strong on floodplains. Storm surge near the coast.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.



GSI “Meiji-era lowland” data, traced from topographic maps of the 1880s–1900s; GSI warns that positions can be off by a considerable distance, especially in the Kanto and Kinki regions. Only low, wet or open ground was traced: towns, dry fields and forest of the period are left blank.

Man-made ground

-  Cut ground 27%
-  Filled and reclaimed land 20%

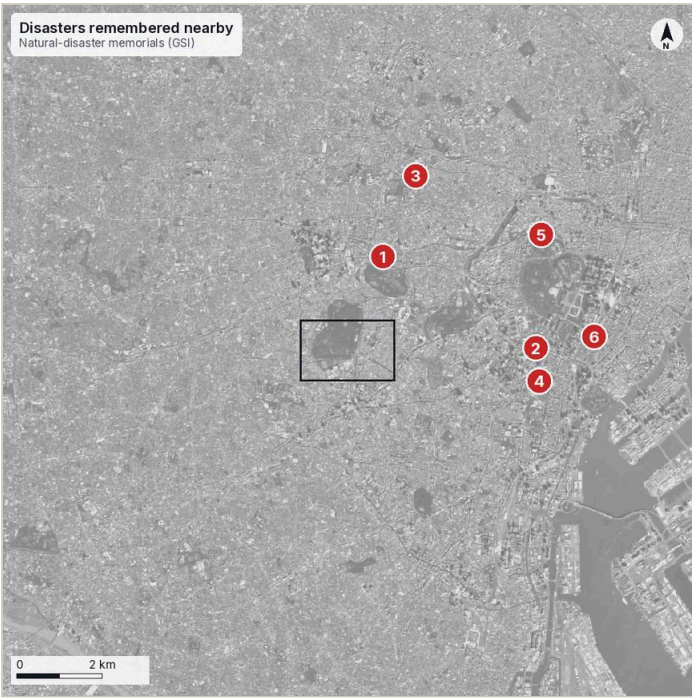
Cut ground (27%). Ground made by cutting into slopes of mountains or terrace edges. Depending on the cut slope, heavy rain or earthquakes may bring it down; the ground itself is generally firm.

Filled and reclaimed land (20%). Ground raised with fill, land made by dumping soil into the sea, rivers or ponds, or valleys filled in. May flood if not raised enough; on hills, rain or earthquakes can make the fill fail; in lowlands the liquefaction tendency is very strong, especially on filled sea, lake or river beds.

Meiji-era lowland

-  Rice paddy 13%
-  River, pond or sea 1.2%

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 16 km across.

Fewer than three memorials stand within 3 km of the study area, so the nearest ones within 8 km are listed.

	Memorial	Disaster	Erected	Distance
1	Memorial to the 1923 Great Kanto Earthquake 大震災記念碑	Great Kanto Earthquake (1 Sep 1923) · earthquake	1929	2.3 km
2	Monument on the site of the Imperial College of Engineering (Kobu Daigakko) 工部大学校址碑	Great Kanto Earthquake (1 Sep 1923) · earthquake	1939	4.4 km
3	Mizukake Jizo statues and memorial to the dead of the 1923 earthquake and fire 水掛け地蔵と大震災火災死亡群霊塔	Great Kanto Earthquake (1 Sep 1923) · earthquake	1924	4.4 km
4	Statue of Sho Kannon in memory of the 1923 Great Kanto Earthquake 震災記念聖観世音菩薩像	Great Kanto Earthquake (1 Sep 1923) · earthquake	1926	4.5 km
5	Monument where Emperor Showa stood to view the city's reconstruction in 1930 昭和天皇御野立所の碑	Great Kanto Earthquake (1 Sep 1923) · earthquake	unknown	5.2 km
6	Memorial for the tenth anniversary of the 1923 Great Kanto Earthquake 関東大震災十周年記念	Great Kanto Earthquake (1 Sep 1923) · earthquake	1933	5.7 km

What the memorials record

- 1. Memorial to the 1923 Great Kanto Earthquake.** In the Great Kanto Earthquake of 1 September 1923, the former Yotsuya Ward had 168 people killed or injured and 500 houses burned. Proposed by newspaper companies in the ward, the memorial was built with the support of 103 backers, including the ward assembly, neighborhood associations and trade associations. Its title was written by the then head priest of Zojoji.
- 2. Monument on the site of the Imperial College of Engineering (Kobu Daigakko).** The Great Kanto Earthquake of 1 September 1923, with more than 100,000 dead and more than 300,000 buildings destroyed or burned, was the largest natural disaster in Japan's history. The former Imperial College of Engineering building here collapsed; the Ministry of Education and the Board of Audit were built in its place. The monument is made from its bricks.
- 3. Mizukake Jizo statues and memorial to the dead of the 1923 earthquake and fire.** In the Great Kanto Earthquake of 1 September 1923 the former Ushigome Ward had 105 people killed or injured and 926 houses wholly or partly destroyed, but almost no fires, so it took in many evacuees. The memorial to the dead was erected the next year; the two standing Jizo statues came from the Gobyō Bridge on Mount Koya.

4. Statue of Sho Kannon in memory of the 1923 Great Kanto Earthquake. Fires broke out in many places after the Great Kanto Earthquake of 1 September 1923, reducing the Shimbashi, Atago and Hamamatsucho areas to ashes. In Shiba Ward, now part of Minato Ward, nearly half of all households were burned out, and the dead were cremated here at Densoin temple.

5. Monument where Emperor Showa stood to view the city's reconstruction in 1930. The Great Kanto Earthquake of 1 September 1923 killed more than 100,000 people, and there were even rumours that the capital would be moved. In 1930, two days before the reconstruction celebrations, Emperor Showa stopped here first while inspecting the rebuilt downtown districts; Tokyo Bay could then be seen from this spot. The stone was rebuilt in 1989 after weathering.

6. Memorial for the tenth anniversary of the 1923 Great Kanto Earthquake. The Great Kanto Earthquake, at 11:58 am on 1 September 1923, left 99,331 dead and 43,476 missing, with 128,226 houses destroyed and 447,128 burned. The slogan carved on the monument, "For the sudden earthquake, constant readiness", urges people to remember the day and never let such a disaster happen again.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Where every element comes from

- **Aerial photographs** (pp. 2–3) — Geospatial Information Authority of Japan (GSI), GSI Tiles: series *ort_riku10* (c. 1936–1942), *ort_USA10* (1945–1950, US military photography), *ort_old10* (1961–1969), *gazo1* (1974–1978), *gazo2* (1979–1983), *gazo3* (1984–1986), *gazo4* (1987–1990) and *nendophoto2019*. Cropped, resized and annotated by Japan Time Atlas. List of GSI Tiles: <https://maps.gsi.go.jp/development/ichiran.html>
- **Landform classification** (pp. 4–5) — GSI landform-classification vector tiles (natural and artificial landforms), compiled by GSI from its Land Condition Maps, Flood Control Terrain Classification Maps and vulnerable-landform surveys. Rendered, measured and translated into English by Japan Time Atlas. https://github.com/gsi-cyberjapan/experimental_landformclassification
- **Meiji-era lowland** (p. 5) — GSI “明治期の低湿地” tiles and legend (https://maps.gsi.go.jp/legend/lw_legend.pdf), measured and translated by Japan Time Atlas.
- **Natural disaster memorials** (p. 6) — GSI Natural Disaster Memorials (自然災害伝承碑), registered by municipalities, <https://www.gsi.go.jp/bousaichiri/denshouhi.html>; processed by Japan Time Atlas (selection by distance, English translation and summary). Memorial photographs are not reproduced.
- GSI content is used under the GSI Content Terms of Use, which follow the Public Data License v1.0 (PDL1.0, compatible with CC BY 4.0). Data retrieved 2026-09-27.
- **Outline of Japan** — Data of Japan / JapanPrefGeoJson (public domain), simplified.

Method

The study area is a fixed rectangle of 2.2 × 1.4 km. Every GSI photo series that covers at least 85% of it is included; the newest annual orthophoto covering at least 90% of it is used as “now”, with any gap filled from GSI's latest seamless mosaic. Landform shares are areas of the rectangle measured on GSI's polygons at zoom level 14, each class merged before measuring. Meiji-era shares count map pixels by legend color. Memorials are selected by straight-line distance from the center of the study area.

Limits

Historical series are mosaics of photographs taken on different dates; GSI publishes the flight date of each photograph. Landform data record conditions at the time of survey. Nothing in this brief shows present-day access, safety or opening hours.

Area Dossier. The same district with a 4–8 page English history, written from Japanese library sources and cited page by page.

Deep Research. A single hamlet or address, researched in libraries and archives.

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